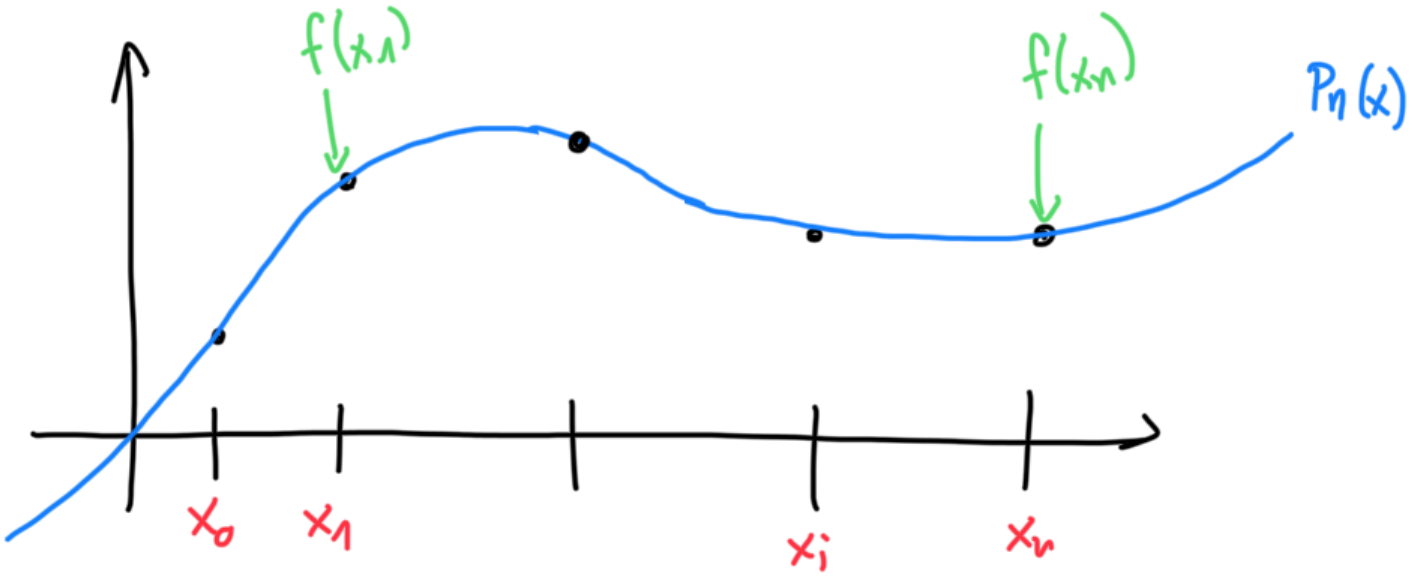
Lecture 14 2025-10-16

Chapter 3: Polynomial Interpolation

Example: Got some data points and want to find a curve going through them, and curve should be a polynomial



In this chapter: Find a polynomial $P_n(x)$ to approximate a function $f(x)$ such that $P_n(x)$ agrees with f at $n+1$ points

$x_0, \dots, x_n \in [a,b]$

i.e. $P_n(x_i) = f(x_i) \quad i = 0, \dots n$

By contrast, the Taylor series for f agrees with the first n derivatives of f at one point x_0.

Recall: Weierstrass' Theorem: If $f \in C([a,b])$ and $\varepsilon > 0$ then $\exists$ a polynomial $p(x)$ s.t. $|p(x) - f(x)| < \varepsilon$ $\forall x \in [a,b]$

Taylor polynomial is only able to satisfy such a condition close enough to x_0.

Example: $f(x) = \log(1+x)$

$P_n(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n+1} \frac{x^n}{n}$

2 3 4 n

(Taylor expansion around $x_0 = 0$)

$$f(2) = \log 3 = 1.0986\ldots$$

n	$P_n(2)$
1	2
5	5.0667
10	−64.8
⋮	⋮

gets worse!

Generalized Rolle's Theorem

Suppose $f \in C^n([a,b])$. If $f(x) = 0$ at $n+1$ distinct numbers $a = x_0 < x_1 < \ldots < x_n = b$, then there exists a $\xi \in (a,b)$ such that $f^{(n)}(\xi) = 0$.

($n = 1$: standard Rolle's theorem)

Proof: On each subinterval $[x_i, x_{i+1}]$, Rolle's Theorem applies, so there exists $\xi_i \in (x_i, x_{i+1})$, $0 \le i \le n$ s.t. $f'(\xi_i) = 0$

Let $x_i^{(1)} = \xi_i$, replace f by f' and repeat...

$f(x)$	has at least	$n+1$	zeros $x_0, \ldots, x_n \in [a,b]$
$f'(x)$	" "	n	zeros $x_0^{(1)}, \ldots, x_{n-1}^{(1)} \in (a,b)$
$f''(x)$	" "	$n-1$	zeros $x_0^{(2)}, \ldots, x_{n-2}^{(2)} \in (a,b)$
⋮		⋮	
(		2	$x_0^{(n-1)} \in (x_0^{(n-2)}, x_1^{(n-2)})$, $x_1^{(n-1)} \in (x_1^{(n-2)}, x_2^{(n-2)})$
$f^{(n)}(x)$		1	zeros $x_0^{(n)} \in (x_0^{(n-1)}, x_1^{(n-1)}) \subset (a,b)$

We note that $f'(x), \ldots, f^{(n-1)}(x) \ldots$ are continuous on (a,b) since they are differentiable, hence they are continuous on each subinterval $[x_i^{(j)}, x_{i+1}^{(j)}]$ for $0 \le i \le n-j$, $j = 0, \ldots, n-1$ ∎

Most straightforward way to find $p_n(x)$:

$$p_n(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$$

$$p_n(x_i) = f(x_i) \qquad i = 0, \ldots, n$$

$$p_n(x_i) = a_0 + a_1 x_i + \ldots + a_n x_i^n = f(x_i) \Leftarrow$$

$$F = (f(x_0), \ldots, f(x_n))^T$$

$$Z = (a_0, \ldots, a_n)^T \quad \text{unknown}$$

$$VZ = F$$

$$V = \begin{pmatrix} 1 & x_0 & x_0^2 & x_0^3 & \ldots & x_0^n \\ 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^n \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^n \end{pmatrix}$$

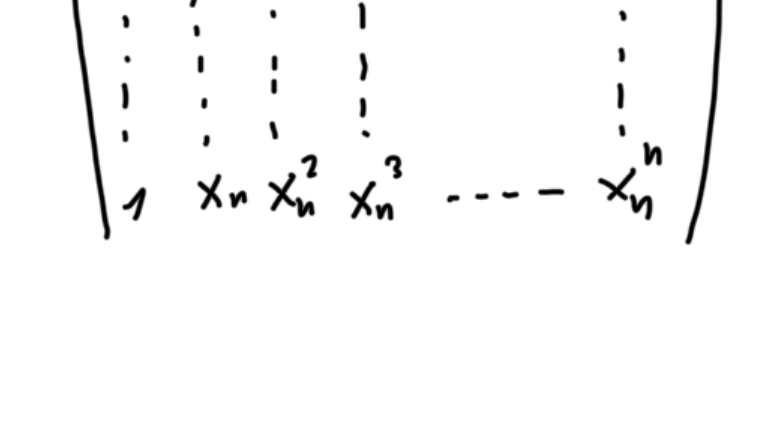
Lagrange Interpolating Polynomial

Given $x_0, \ldots x_n$, we define $L_k(x) = \underbrace{\prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{x - x_j}{x_k - x_j}}$ $\quad 0 \le k \le n$

$$\frac{x-x_0}{x_k-x_0} \cdot \frac{x-x_1}{x_k-x_1} \cdots \frac{x-x_{k-1}}{x_k-x_{k-1}} \frac{x-x_{k+1}}{x_k-x_{k+1}} \cdots \frac{x-x_n}{x_k-x_n}$$

L_k is a polynomial of degree n

$$L_k(x_j) = \delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$



Example: Let $\{x_i\} = [\overset{x_0}{1}, \overset{x_1}{2}, 5, 6, \overset{x_4}{7}]$ $\quad n = 4$

$$L_1(x) = \left(\frac{x-1}{2-1}\right)\left(\frac{x-5}{2-5}\right)\left(\frac{x-6}{2-6}\right)\left(\frac{x-7}{2-7}\right)$$

$L_1(1) = 0 \qquad L_1(5) = 0,\ L_1(6) = 0,\ L_1(7) = 0$

$L_1(2) = 1$

Definition: Given $f(x)$ and nodes $\{x_k\}_{k=0}^n$, the Lagrangian interpolating polynomial of f is

$$P(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$$

Note: $P(x)$ is a polynomial of degree n and

$$P(x_j) = \sum_{k=0}^{n} f(x_k) \underbrace{L_k(x_j)}_{\delta_{jk}} = f(x_j)$$

Theorem: (analog of Taylor's theorem with remainder)

Suppose $f \in C^{n+1}([a,b])$ and $x_0, \dots, x_n \in [a,b]$ are distinct.

Then $\forall x \in [a,b]$ there exists $\xi(x) \in (a,b)$ such that

$f(x) = P(x) + R(x)$ with

$$P(x) = \sum_{k=0}^{n} f(x_k) L_k(x) \quad \text{and}$$

$$R(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n)$$

Proof: 1) If $x = x_k$, $f(x_k) = P(x_k)$ and $R(x_k) = 0$ (for any choice of ξ)

2) Otherwise let $x \in [a,b]$ (not equal to x_k)

$$g(t) = f(t) - P(t) - [f(x) - P(x)] \prod_{j=0}^{n} \frac{t - x_j}{x - x_j}$$

$t = x$

- $g(x) = 0$ (plug in & check)

$t = x_k$

- $g(x_k) = 0 \quad k = 0, \dots, n$ (plug in & check)

(recall $P(x_k) = f(x_k)$ & $\prod_{j=0}^{n} \cdots \Big|_{t = x_k} = 0$)

$\Rightarrow$ g has $n+2$ zeros.

$\Rightarrow$ Generalized Rolle's theorem implies that $\exists\ \xi \in (a,b)$

s.t. $g^{(n+1)}(\xi) = 0$

$$\frac{d^{n+1}}{dt^{n+1}} P(t) = 0 \qquad (P \text{ is of degree } n)$$

$$0 = g^{(n+1)}(\xi) = f^{(n+1)}(\xi) - [f(x) - P(x)] \frac{d^{n+1}}{dt^{n+1}} \left(\prod_{j=0}^{n} \left(\frac{t - x_j}{x - x_j} \right) \right)$$